SIGNAL AND POWER INTEGRITY: Let Your Product Be In Harmony

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An LED light manufacturing unit was working on a 150-watt streetlight project wherein the engineering design team had to rack their brains for a long time over an issue. Their LED supply boards were malfunctioning because the controller ICs would burn out within a few seconds. Later the team realised that the ICs were faulty. The IC manufacturer had to confess about the signal losses arising from the use of poor-quality controller IC. The internal built of the IC was not reliable and had issues with the layout. The entire issue came to light only after the ICs were tested on a signal integrity (SI) measurement instrument.

Likewise, designers have reported component failures when these were not tested with power integrity (PI) measurement tools.

Need for signal and power integrity

While measuring critical parameters like drain-to-source voltage of MOSFETs, design engineers face frequency fluctuations or varying time-periods as a result of signal integrity issues. These occur due to ground bounce or poor signals, or due to signal losses and power supply noise. Similarly, each electrical system has its own impedance, which can be impacted due to power integrity issues. Thus, to get the benefits of clean data, time saving and more reliable products, signal and power integrity instruments become a necessity.

Fig. 1: Ringing effect in the coil (in magenta colour) due to signal integrity issues (Image courtesy: http://jeelabs.org)

Market overview

In late 1990s signal integrity issues were common, so designers focused on traces and path of the PCB design too. And now the power supplied to circuits has also become a matter of concern as it gives rise to power integrity issues.

Today, designers work on high-speed designs, where signal integrity and power integrity analyses are equally important. Power integrity issues in a design can actually occur as signal integrity issues. As designs become increasingly compact, design engineers are facing far too many integrity issues, which has created a demand for constant upgrades in measurement tools.

Troubleshooting with MSO

Signal integrity issues in analogue domain include impedance mismatch, crosstalk and transmission-line effects, while digital issues are setup and hold violation.

As analogue signals are responsible for digital signal mismatch, mixed-signal oscilloscopes (MSOs) have been introduced in the market that give a combined result of analogue and digital signals. Thus, a designer can monitor several points in a design simultaneously.

MSOs are a powerful tool for troubleshooting signal integrity problems.

Enhancing probes to increase the accuracy

Companies have also introduced low-current probes in oscilloscopes to improve the measurement accuracy. These upgrades focus on low-current measurements for small applications such as LED drivers and now feature 1GHz frequency.